

NetComms 2: Summative Assessment 1 Reviewer

Modules 1 and 2, DevNet Associate. Built from DEVASC_Module_1.pdf and DEVASC_Module_2.pdf.

NetComms 2 Module 1: Course Introduction and DevNet Lab Environment

Course Overview

The DevNet Associate (DEVASC) course introduces software development, networking fundamentals, and automation. It is organized into five modules covering APIs, software design, network fundamentals, infrastructure automation, and Cisco platforms.

Key terms

- **DevNet Associate** — Cisco certification focusing on software development and automation for network engineers.
- **API** — Application Programming Interface; a set of rules that allows software components to communicate.
- **Automation** — Using software tools to perform tasks without manual intervention.

Lab Environment Setup

Students must install a virtual lab environment using a hypervisor. The virtual machines (VMs) act as guests on the physical host computer, providing a consistent Linux-based platform for all labs.

Key terms

- **Virtual Machine (VM)** — A software emulation of a physical computer that runs its own operating system.
- **Host** — The physical computer that runs the hypervisor and hosts VMs.
- **Guest** — The virtual machine that runs inside the host.

Key points

- **Isolation:** Each VM is isolated from the host and other VMs, preventing interference between lab environments.
- **Portability:** VM images can be exported and imported across different host hardware.

Example. Create a new Ubuntu VM for the DEVASC labs.

1. Open your hypervisor (e.g., VirtualBox or VMware).
2. Click 'New' and name the VM 'DEVASC_Ubuntu'.
3. Allocate at least 2 GB RAM and 20 GB disk space.
4. Mount the Ubuntu ISO as the optical drive.
5. Start the VM and follow the Ubuntu installation prompts.
6. Install the Guest Additions/Tools for better integration.

Answer: A fully functional Ubuntu VM named DEVASC_Ubuntu ready for lab work.

Watch out for

- Forgetting to enable hardware virtualization (VT-x/AMD-V) in the BIOS, causing the VM to fail to start.
- Allocating too little RAM, leading to sluggish performance during lab exercises.

Linux Fundamentals for DevNet

All coding labs are performed on a Linux-based VM. The module reviews Linux file system navigation, file permissions, regular expressions, and basic system administration.

Key terms

- **File System Hierarchy** — Standard directory layout in Linux (/, /home, /etc, /var, /usr, etc.).
- **Permissions** — Read, write, and execute rights for user, group, and others (rwx).
- **Regular Expression** — A pattern that describes a set of strings, used for searching and text processing.

Key points

- **Permission Bits:** Each file/directory has a 9-bit permission set: rwx for owner, group, and others.
- **Root Privileges:** The root user can override all permission restrictions.

Example. Change a file's permissions so that the owner can read/write, the group can read, and others have no access.

1. Identify the file, e.g., myscript.sh.
2. Use the chmod command with symbolic mode: `chmod u=rw,g=r,o= myscript.sh`
3. Verify with `ls -l myscript.sh`.

Answer: `-rw-r----- 1 user group 0 Sep 14 12:00 myscript.sh`

Example. Search for lines containing an IP address pattern in a log file using a regular expression.

1. Use grep with a regex: `grep -E '[0-9]{1,3}(\.[0-9]{1,3}){3}' syslog.txt`
2. Review the output for matching lines.

Answer: Displays all log lines that contain an IPv4 address.

Watch out for

- Using numeric chmod values without understanding the bit mapping, leading to overly permissive settings.
- Running commands that modify system files without sudo, resulting in permission denied errors.

Python Programming Basics

Python is introduced as an easy-to-learn, open-source language that serves as a foundation for other languages. Labs cover data types, variables, lists, dictionaries, control structures, file I/O, and the input function.

Key terms

- **Variable** — A named reference to a value stored in memory.
- **List** — An ordered, mutable collection of items, defined with square brackets []
- **Dictionary** — A collection of key-value pairs, defined with curly braces {}
- **Control Flow** — Structures like if, for, and while that direct execution order.

Key points

- **Dynamic Typing:** Python determines variable types at runtime; no explicit type declarations are required.
- **Indentation Sensitivity:** Block structure is defined by consistent indentation (usually 4 spaces).

Example. Write a Python script that reads a list of numbers from the user, calculates the average, and writes the result to a file called avg.txt.

1. Prompt the user for a comma-separated string: `nums = input('Enter numbers: ')`
2. Convert the string to a list of integers: `num_list = [int(x) for x in nums.split(',')]`
3. Compute the average: `avg = sum(num_list) / len(num_list)`
4. Open a file for writing: with `open('avg.txt', 'w')` as `f`:
5. Write the result: `f.write(f'Average: {avg}\n')`
6. Print confirmation to the console.

Answer: The file avg.txt contains a line such as 'Average: 23.5' based on user input.

Example. Demonstrate a `for` loop that prints the squares of numbers 1 through 5.

1. Use a range iterator: `for i in range(1, 6):`
2. Inside the loop, `print(i**2)`.

Answer: Output: 1 4 9 16 25 (each on its own line).

Watch out for

- Mixing tabs and spaces, causing `IndentationError`.
- Forgetting to convert input strings to integers before arithmetic, leading to `TypeError`.

Module Summary and Self-Assessment

The Course Introduction module prepares learners for the DEVASC program by ensuring they have a functional virtual lab, basic Linux proficiency, and introductory Python skills. Successful completion of the labs confirms readiness for subsequent modules.

NetComms 2 Module 2: The DevNet Developer Environment

DevNet Overview

DevNet is Cisco's integrated developer program that offers a website, community, tools, forums, sandboxes, and documentation to help network programmers build, test, and share solutions using Cisco APIs.

Key terms

- **DevNet** — Cisco's developer program that provides resources, tools, and community support for building network automation and programmability solutions.
- **API** — Application Programming Interface; a set of rules that allows software applications to communicate with Cisco platforms.
- **Sandbox** — A production-like, isolated environment where developers can safely experiment with Cisco technologies without affecting live networks.

Key points

- **Learning Labs:** Self-paced tutorials covering coding basics to advanced REST API usage.
- **Code Exchange:** Repository of sample code contributed by the community, searchable via tags and categories.
- **Developer Support:** Ticketing, live chat, and forum assistance for integration and API questions.

Example. Locate the DevNet Home page and identify the main navigation tiles.

Open a web browser and go to <https://developer.cisco.com>.

Observe the homepage layout; the main tiles are: Start Now, Learning Tracks, Video Course, Sandbox, Code Exchange, Ecosystem Exchange.

Click each tile to verify it leads to the described resource.

Answer: The homepage displays six primary tiles that serve as entry points to the core DevNet resources.

Watch out for

- Assuming the sandbox is a production environment – it is isolated and resets after each session.
- Skipping the Learning Labs and jumping straight to code samples, which can lead to misunderstanding API authentication flows.

Exploring DevNet Online Resources

DevNet aggregates multiple online assets—Learning Labs, Sandboxes, Exchanges, and support portals—allowing developers to discover tutorials, test environments, reusable code, and community solutions.

Key terms

- **Learning Tracks** — Curated paths that group related labs and videos to guide developers from beginner to advanced topics.
- **Exchange** — A category of community-driven repositories (Automation, Code, Ecosystem) that share use cases, scripts, and solution designs.

Example. Find a use case for network automation in the Automation Exchange.

Navigate to <https://developer.cisco.com/automation-exchange>.

Use the filter panel to select a technology (e.g., Ansible) and a scenario (e.g., Campus).

Open a listed use case, read the description, and note the provided playbooks.

Answer: A sample use case titled "Campus VLAN provisioning with Ansible" appears, offering a ready-to-run playbook and step-by-step guide.

Watch out for

- Choosing a sandbox that does not match the intended platform (e.g., selecting a data center sandbox for a branch router use case).
- Overlooking the Ecosystem Exchange, which can contain prebuilt integrations for third-party tools.

DevNet Learning Labs

Learning Labs are self-paced, interactive tutorials that cover programming languages, model-driven programmability, REST APIs, and hands-on sandbox usage, enabling developers to set up local environments and practice real-world scenarios.

Key terms

- **REST API** — Representational State Transfer API; a web-service interface that uses HTTP methods to manipulate resources.
- **Model-Driven Programmability** — An approach where network behavior is defined by data models (YANG) that can be programmatically accessed via APIs.

Key points

- **Hands-on Sandbox Integration:** Each lab includes steps to connect to a pre-configured sandbox for immediate testing.
- **Multi-language Support:** Labs are available for Python, JavaScript, and other common scripting languages.

Example. Complete the "Python REST API" lab to retrieve interface details from a sandbox router.

Launch the lab from the Learning Labs catalog.

Follow the lab instructions to install the required Python packages (requests, json).

Copy the sandbox URL and authentication token provided in the lab.

Write a Python script that sends a GET request to /restconf/data/ietf-interfaces:interfaces.

Run the script and verify the JSON output lists all interfaces.

Answer: The script returns a JSON object containing interface names, statuses, and IP addresses from the sandbox router.

Watch out for

- Running lab scripts without activating the associated sandbox first, resulting in connection errors.
- Ignoring the lab's prerequisite software versions, which can cause import failures.

DevNet Sandbox

Sandboxes provide on-demand, production-like environments for a variety of Cisco technologies (e.g., IOS-XE, NX-OS, DNA Center). They enable developers to test APIs, configurations, and automation scripts safely.

Key terms

- **Sandbox Reservation** — The process of claiming a sandbox instance for a limited time window (typically 2-4 hours).

- **Reset Cycle** — Automatic restoration of the sandbox to its original state after the reservation expires.

Key points

- **Time■Limited Access:** Each sandbox reservation lasts a predefined period; after expiration the environment resets.
- **Multiple Topologies:** Available sandboxes include single■device, multi■device, and full■stack solutions.

Example. Reserve a Cisco DNA Center sandbox and retrieve the system health via its REST API.

Log in to developer.cisco.com and navigate to the Sandbox catalog.

Select the "Cisco DNA Center" sandbox and click "Reserve".

Copy the provided base URL, username, and password.

Use a tool like Postman to send a GET request to `/api/v1/system/health`.

Inspect the JSON response for health metrics such as CPU, memory, and service status.

Answer: The API returns a JSON payload showing all health indicators as "normal" for the newly reserved sandbox.

Watch out for

- Attempting to use a sandbox after the reservation window has closed, leading to authentication failures.
- Modifying the sandbox configuration and expecting changes to persist across sessions; they are cleared on reset.

DevNet Exchanges

Exchanges are community■driven portals that host automation use cases, reusable code, and solution designs. The three primary exchanges are Automation, Code, and Ecosystem.

Key terms

- **Automation Exchange** — A collection of network automation use cases, scripts, and toolkits (e.g., Ansible, Puppet).
- **Code Exchange** — A searchable repository of sample code and tools, indexed via the GitHub API and moderated by Cisco staff.
- **Ecosystem Exchange** — A marketplace of 1,500+ solutions spanning multiple technologies, industries, and geographies.

Example. Locate a Python script that automates VLAN creation on a Cisco Catalyst switch via the Code Exchange.

Visit <https://developer.cisco.com/code-exchange>.

Enter keywords "VLAN" and "Python" in the search bar.

Filter results by "Cisco IOS XE" platform.

Select a repository titled "cisco■vlan■automation".

Review the README for usage instructions and copy the script.

Answer: The selected repository provides a Python script that uses the RESTCONF API to create, modify, and delete VLANs on a Catalyst switch.

Watch out for

- Downloading code without checking the compatibility matrix (e.g., using an IOS■XE script on an NX■OS device).
- Neglecting to review licensing information; some code may be under restrictive licenses.

DevNet Developer Support

Developer Support offers multiple channels—tickets, community forums, WebEx Teams, and a knowledge base—to help troubleshoot API integrations, connectivity issues, and other developer■focused questions.

Key terms

- **Support Case** — A ticket submitted through the DevNet portal that receives a response within one business day.
- **Knowledge Base** — A curated collection of troubleshooting articles and best■practice guides for developers.

Key points

- **Response Time:** Standard support cases are answered within one business day; higher■tier support may have faster SLAs.
- **Community Forums:** Public forums where developers can post questions and receive answers from peers and Cisco engineers.

Example. Open a support ticket for an authentication error when calling a Cisco DNA Center API.

Navigate to <https://developer.cisco.com/support>.

Click "Create a Case" and fill in the required fields: product (DNA Center), API endpoint, error message, and steps to reproduce.

Attach a snippet of the failing request (redacted credentials).

Submit the case and note the case number.

Monitor email for the Cisco response, which typically includes token generation guidance.

Answer: The support team replies with a corrected authentication flow, resolving the error.

Watch out for

- Providing insufficient detail in a support case, leading to back and forth clarification and delayed resolution.
- Posting sensitive credentials in public forums; always redact or use placeholders.

Lab – Explore DevNet Resources

The hands-on lab reinforces navigation of Learning Labs, Sandboxes, and Exchanges. It consists of two parts: locating a Learning Lab and exploring additional resources such as video courses and the Automation Exchange.

Key terms

- **Lab Part 1** — Task to find and launch a specific Learning Lab from the DevNet catalog.
- **Lab Part 2** — Task to browse other DevNet resources (video courses, sandboxes, exchanges) and document findings.

Example. Complete Lab Part 1: Find the "REST API Basics" Learning Lab and start the associated sandbox.

Log in to developer.cisco.com.

Select "Learning Labs" from the main navigation.

Search for "REST API Basics" and click the lab title.

Follow the lab instructions to reserve the linked sandbox.

Verify connectivity by executing the provided curl command.

Answer: The sandbox is successfully reserved, and the curl command returns a JSON list of interfaces, confirming lab completion.

Example. Complete Lab Part 2: Identify a video course that covers network automation with Ansible.

From the DevNet homepage, click "Video Course".

Use the filter to select "Automation" and "Ansible".

Select the course titled "Automating Cisco Networks with Ansible".

Note the course duration, prerequisites, and enrollment link.

Answer: The video course is listed with a 2-hour runtime and requires basic Python knowledge; enrollment is open.

Watch out for

- Skipping the sandbox reservation step, which causes the lab's API calls to fail.
- Assuming the video course is automatically enrolled; users must click the enrollment button.

Self-check

Answer these from memory before looking at the key below.

- 1 What is the primary purpose of installing a virtual lab environment for the DEVASC course?
- 2 Explain the difference between a host and a guest in virtualization terminology.
- 3 List three Linux file permission sets and describe what each letter (r, w, x) stands for.
- 4 Write a one-line Python expression that creates a dictionary mapping the numbers 1-3 to their squares.
- 5 What common mistake can cause a Python script to raise an IndentationError?
- 6 What are the three primary components of the DevNet Exchange ecosystem?
- 7 Describe two key differences between a DevNet Learning Lab and a Sandbox.
- 8 If you encounter a 401 Unauthorized error when calling a Cisco API from a sandbox, what support options are available?
- 9 How long does a typical DevNet sandbox reservation last, and what happens after it expires?
- 10 List two common mistakes developers make when using the Code Exchange.

Answer key

- 1 To provide a consistent, isolated Linux-based platform where all coding labs can be executed without affecting the host system.

- 2 The host is the physical computer that runs the hypervisor; the guest is the virtual machine that runs its own operating system inside the host.
- 3 r = read permission, w = write permission, x = execute permission. The three sets apply to owner, group, and others respectively.
- 4 `{i: i**2 for i in range(1, 4)}`
- 5 Mixing tabs and spaces or using inconsistent indentation levels within the same block.
- 6 Automation Exchange, Code Exchange, and Ecosystem Exchange.
- 7 Learning Labs are tutorial-driven, step-by-step guides that teach concepts; Sandboxes are isolated, production-like environments used to test configurations and APIs. Labs often include a sandbox reservation as part of the exercise, but the sandbox itself does not provide instructional content.
- 8 You can log a Support Case (ticket) via the DevNet support portal, post the issue to the community forums, or start a WebEx Teams chat with the support team. Additionally, consult the Knowledge Base for authentication troubleshooting articles.
- 9 Reservations usually last 2-4 hours. After expiration, the sandbox automatically resets to its original factory state, discarding any configuration changes made during the session.
- 10 1) Downloading code without verifying platform compatibility (e.g., using an IOS-XE script on an NX-OS device). 2) Ignoring licensing terms, which may restrict commercial use of the sample code.